

Mithilesh Chaurasiya

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Professional Summary

Machine Learning Engineer with a strong background in data engineering and MLOps, bringing 2.8 years of experience from Infosys. Skilled in building end-to-end machine learning pipelines, including data ingestion, model training, deployment, and monitoring using tools like Airflow, MLflow, and DVC. Experienced in developing real-time Computer Vision systems and optimizing data workflows for scalability and performance. Passionate about solving real-world problems through production-grade ML systems.

Technical Skills

- **Programming:** Python, Java
- **Machine Learning & Deep Learning:** Scikit-learn, PyTorch, CNNs, Tensorflow
- **Computer Vision:** OpenCV, YOLO, Object Detection
- **Generative AI:** Langchain, Agentic AI
- **MLOps & Deployment:** MLflow, DVC, Apache Airflow, Docker, Kubernetes (Minikube),
- **Data Engineering:** Apache Spark, MongoDB, Snowflake, Hadoop
- **Data Analysis:** Pandas, NumPy, SQL (Oracle)
- **Cloud:** Microsoft Azure
- **Automation:** Selenium, UiPath
- **Tools & Version Control:** Git, GitHub
- **Web Development:** Flask
- **FastAPI, CI/CD (Jenkins, Azure DevOps)**

Core Strengths

- **Problem Solving:** Strong analytical approach to optimizing data pipelines and ML workflows.
- **Leadership & Mentorship:** Leading the Machine Learning team at DCC, mentoring students on ML, Computer Vision, and real-world project development.
- **Collaboration:** Experience working in Agile teams with cross-functional stakeholders on data-driven systems.
- **Communication:** Ability to translate business requirements into scalable technical solutions.

Education

Master of Computer Application (MCA) 2024 – Expected 2027
National Institute of Technology, Agartala

Bachelor of Science in Computer Science 2021
Thakur College of Science and Commerce, Mumbai
CGPA: 8.87/10

Higher Secondary Certificate (HSC) 2018
Utkarsha Vidyalaya & Junior College, Mumbai
Percentage: 64%

Secondary School Certificate (SSC) 2016
Jai Deep Vidya Mandir High School, Mumbai

Percentage: 75%

Professional Experience

Infosys Ltd. June 2021 – January 2024
Senior System Associate (Data Engineering & ETL Testing)

- Validated and monitored production-grade ETL pipelines handling large-scale structured data across Oracle, MongoDB, and Snowflake, ensuring high data integrity for downstream ML and analytics systems.
- Built Python-based automation frameworks for data validation and API testing, reducing manual QA effort by 30%+ and improving overall pipeline reliability and efficiency.
- Optimized ETL workflows to generate ML-ready datasets, improving data availability and reducing pre-processing time for analytics and machine learning use cases.
- Performed end-to-end validation of REST APIs and model endpoints, ensuring correctness and consistency of model outputs in integrated production systems.
- Collaborated with cross-functional teams to implement ML experiment tracking (MLflow) and dataset versioning (DVC), improving reproducibility, traceability, and consistency of model development workflows.
- Worked in Agile/Scrum environments, contributing to sprint planning, testing strategies, and client-facing deliverables in a cross-functional team setting.

Projects

1. Smart Traffic Management System (End-to-End CV + MLOps Project)

YOLO (Ultralytics), OpenCV, Airflow, MLflow, DVC, MongoDB, FastAPI, Streamlit

- Built a production-grade computer vision + MLOps system for automated traffic analytics.
- Implemented real-time vehicle detection and multi-object tracking for persistent ID assignment.
- Designed traffic intelligence modules for vehicle count, density, and flow estimation.
- Orchestrated complete ML lifecycle (data → training → inference) using Apache Airflow DAGs.
- Integrated DVC for large-scale dataset versioning (22GB IDD dataset) ensuring full reproducibility.
- Implemented MLflow-based experiment tracking with dataset hash → model version traceability.
- Added training deduplication logic to automatically skip duplicate experiments, reducing GPU cost.
- Built Streamlit dashboard for real-time traffic monitoring and analytics visualization.
- Designed modular, production-ready architecture following real-world ML system design principles.

2. Stock Market ETL Pipeline & Dashboard

Python, Airflow, MongoDB, Streamlit

- Built a complete ETL pipeline to fetch and clean stock market data, storing it in MongoDB.

- Automated daily workflows using Apache Airflow.
- Developed an interactive Streamlit dashboard showing stock price trends, SMA, and volume.
- Optimized MongoDB queries using indexing and caching for faster performance.

3. Real-Time Gender Detection

Python, TensorFlow, OpenCV, CNN

- Developed a CNN-based system to classify gender from facial images with over 90% accuracy.
- Implemented real-time face detection using OpenCV.
- Fine-tuned hyperparameters to optimize model performance.

4. MockAnalysis: Automated Mock Test Result Analysis System

Python, Selenium, Pandas, Matplotlib

- Automated scraping and analysis of mock test results.
- Visualized insights with Pandas and Matplotlib; reduced manual effort by 40%.
- Hosted the project on GitHub for portfolio demonstration.

Heart Disease Prediction ML Model

Python, Scikit-learn, Pandas, Matplotlib

- Built a machine learning model to predict heart disease using patient records.
- Performed feature engineering, classification, and performance evaluation.

Leadership & Extracurricular Activities

Machine Learning Team Lead — Developers and Coders Club (DCC)

National Institute of Technology, Agartala

- Leading the Machine Learning team responsible for mentoring junior students in ML, AI, and Data Science.
- Conducted sessions on Machine Learning fundamentals, Deep Learning, and Computer Vision.
- Guided students in building real-world ML projects and Kaggle-based learning.
- Coordinating with faculty mentors on research-oriented and project-based initiatives.
- Planning technical workshops, roadmap sessions, and ML awareness programs for NIT Agartala students.
- Volunteer, College Fest Committee — Assisted in planning and organizing technical events and cultural activities.

Certifications

- Accenture North America - Data Analytics and Visualization Job Simulation
- Infosys Certified Selenium with Python Automation Tester
- Infosys Certified Software Development Engineer in Test (SDET)
- Infosys Certified Digital Assurance Professional
- Infosys Certified Python Programmer
- Infosys Certified Data Management Associate
- Machine Learning Foundations — Great Learning
- Computer Vision Essentials — Great Learning
- Python for Machine Learning — Great Learning

Hobbies

- Reading
- Photography
- Traveling